ERP Report

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Roll Number:	FT272022
Course:	ERP and Industry 4.0
Chosen Process:	Manufacturing (Plan to Produce)
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Business Context

Organization Overview

Premium Auto Ltd. is a luxury automobile manufacturer headquartered in India, specializing in high-end sedans and SUVs for Indian and Southeast Asian markets. The company operates with approximately 2,500 employees across manufacturing, sales, and support functions. Annual production capacity exceeds 15,000 units, with rigorous quality standards aligned with international automotive norms.

Current Process

The Plan-to-Produce (P2P) process orchestrates the journey from sales order to production completion. This encompasses demand planning, inventory management, procurement coordination, production scheduling, and quality verification. The process currently involves multiple manual touchpoints across departments, resulting in delays and potential errors.

Business Problem

The existing manual-heavy approach creates several bottlenecks. Production schedules conflict with inventory availability. Supplier coordination happens through email chains, leading to communication gaps. Quality discrepancies are discovered late in production. Delays cascade through the supply chain. There is no real-time visibility into process status, making it difficult to meet delivery timelines for high-value orders. Decision-making at critical gateways such as supplier availability and quality acceptance lacks standardized criteria.

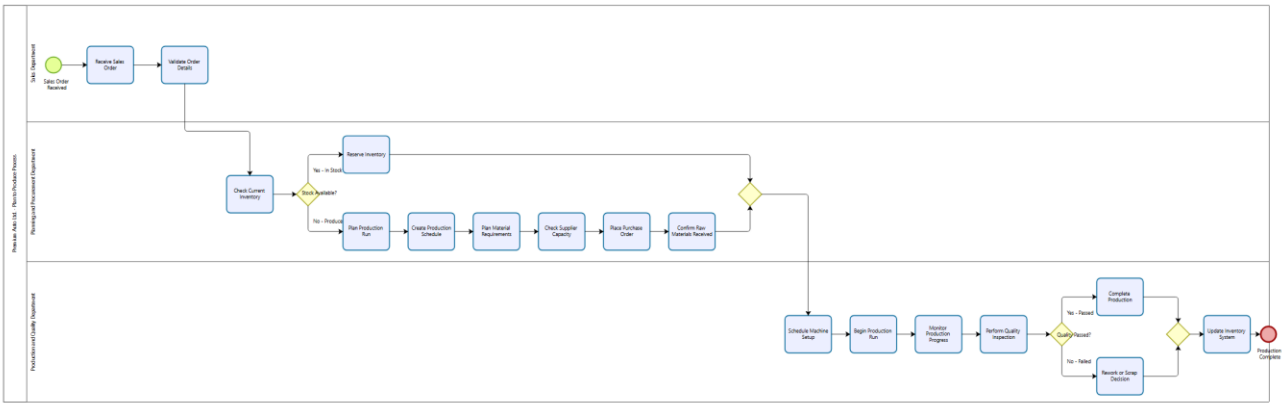
Proposed Solution

The ERP rollout closes the gaps that cause most of the current delays. The order-to-planning handoff moves from email into a single ERP workflow, cutting the 3-4 day lag before MRP picks up new orders. Approvals at the two decision gateways (stock check and quality inspection) sit inside the system with a fixed audit trail, so escalations don't stall waiting on someone to reply. Alongside the ERP, ServiceNow handles the exception workflows that currently live in email — the Finance approval for POs above ₹25 lakh, capacity conflict escalations to VP Operations, and the rework-or-scrap decisions after quality failures — with an SLA on each step and one case record that follows the exception across departments.

Process Scope

Scope Element	Definition
Start Event	Sales Order received from customer (order confirmation with delivery date and specifications)
End Event	Production complete and order ready for shipment (vehicle passes final QC and is updated in inventory)
Process Trigger	Customer places confirmed order through sales portal with deposit received
Expected Output	Manufactured vehicle meeting quality standards, updated in ERP, ready for logistics and delivery
Primary Stakeholder	Production Manager (owns end-to-end process execution and timeline adherence)
Supporting Stakeholders	Sales Team, Planning Department, Procurement, Quality Assurance, Finance (approval authority for supplier payments)

BPMN Diagram



Business Rules

The following business rules govern decision points and validations within the FT272022_NamanJain_PlanToProduce.xlsx

BR ID	Business Rule	Context / Impact
BR-04	Available units must be hard-allocated to the order (reserved specifically, not soft-reserved); production planning branch is skipped	Expedites delivery for in-stock variants; reduces lead time
BR-06	Production planning branch is initiated with a minimum lead time of 45 days; compression below 45 days requires VP Operations approval	Ensures realistic production schedules; avoids over-committing capacity
BR-11	PO with value ≥ ₹25 lakh must be routed to Finance for approval before dispatch to the supplier	Controls spend authority; ensures budget compliance
BR-10	Supplier capacity confirmation must be received within 5 business days of PO release; day-5 default triggers alternate supplier engagement	Prevents delays in material procurement; maintains supply chain rhythm
BR-15	Quality inspection sample size must be the higher of 10% of the batch or 5 units; sampling below this threshold is not permitted	Statistically valid quality sampling; compliance with automotive standards
BR-17	Rework is permitted up to 2 iterations per unit; a third attempt is blocked in the system and the unit is routed to scrap	Balances quality with cost; prevents infinite rework cycles for defective units
BR-08	Schedule drafts exceeding available machine or labour capacity raise an exception to Production Manager and VP Operations; not released until approved in writing	Escalates bottlenecks to management; prevents unrealistic commitments
BR-20	Projected dispatch beyond +7 days from the promised delivery date requires written customer approval before the order can proceed	Protects customer satisfaction and brand reputation

Process Improvement Suggestions

Based on process analysis, the following three improvements are recommended for implementation during ERP rollout.

1. Parallel Supplier Confirmations & Automated Notifications:

Instead of sequential supplier contact (current state), send purchase orders to 3-4 pre-qualified suppliers in parallel within the same hour of order receipt. Each supplier receives an automated notification with a 24-hour SLA for capacity confirmation. The first supplier to confirm capacity is locked in; others are automatically notified of cancellation.
Impact: Reduces supplier confirmation time from 5-7 days to 1-2 days, accelerates production scheduling, and improves on-time delivery rate.

2. Automated Exception Routing via ServiceNow:

Route the four exception paths — Finance approval for POs above ₹25 lakh, supplier confirmation SLA breaches, capacity conflicts flagged during scheduling, and rework-or-scrap decisions after quality failures — through ServiceNow instead of email chains. Each exception opens a case with a named owner, an SLA clock, and a fixed escalation ladder (owner at 24 hours, manager at 48 hours, VP at 72 hours). The case record travels with the exception across departments, so nobody has to reconstruct the history when it lands on the next desk.
Impact: Cuts the average exception resolution time from 4-5 days to under 48 hours, and removes the recurring problem of exceptions sitting unanswered in someone's inbox.

3. Inventory Buffer & Production Levelling with Demand Forecasting

Maintain a rolling 30-day demand forecast (updated weekly) to pre-plan production for high-volume, fast-moving variants. Build a strategic buffer stock (5-10% of monthly demand) for popular models to reduce make-to-order lead times. Create production levelling rules that batch similar orders to reduce changeover time and improve machine utilization. Use ERP dashboards to provide customers with real-time delivery date confirmation instead of static estimates.
Impact: Improves on-time delivery predictability, reduces customer disappointment, and increases production efficiency by 15-20%.